

Table 1: Variant comparison: the five HypatiaX hybrid variants (V1–V5) against each other and against Neural Network (NN), pure LLM and PySR baselines. Each panel is one benchmark on which the compared methods share the same equations; panels are *not* comparable with one another. Every row is tied to source code by a check run at generation time (see `variant_comparison_evidence.json`); variants with no usable result source are listed in panel D with the evidence rather than given placeholder numbers.

(A) Feynman-30, in-distribution R^2 (noiseless protocol, $n = 30$ equations; pass = $R^2 \geq 0.99$)						
Role	Recorded method label	ok/n	Median R^2	Pass	Median time (s)	
LLM	PureLLM Baseline (core)	29/30	1.0000	29/30	6.0	
NN	ImprovedNN (core)	30/30	0.9999	30/30	0.6	
V1	EnhancedHybridSystemDeFi (core)	30/30	1.0000	30/30	9.9	
V2	HybridSystemLLMNN all-domains (core)	30/30	1.0000	30/30	8.9	
V4 (llm_mode=none, LLM off)	SymbolicEngineWithLLM (tools)	30/30	1.0000	30/30	37.3	
PySR engine (v50.2, no LLM)	HybridDiscoverySystem v50.2 (tools)	30/30	1.0000	30/30	61.4	

(B) Core-15, extrapolation R^2 (median over successful runs; (k) = number of the n equations with a finite value)						
Role (source row)	ok/n	Train	Near (k)	Med (k)	Far (k)	Time (s)
NN (<i>Neural Network</i>)	15/15	0.9976	-0.1929 (14)	-0.4931 (14)	-14.7408 (12)	0.5
LLM (<i>Pure LLM</i>)	15/15	0.9930	-1.4868 (1)	— (0)	— (0)	7.1
PySR-only (<i>exp1_ablation pysr-only (raw PySRRegressor)</i>)	15/15	0.9975	1.0000 (15)	1.0000 (14)	1.0000 (15)	1101
V2 (<i>System 3 LLM+Fallback</i>)	15/15	1.0000	— (0)	— (0)	— (0)	9.3
V4 (llm_mode=none, LLM off) (<i>System 2 Symbolic</i>)	15/15	0.9978	0.9987 (12)	0.9969 (12)	0.9461 (12)	33.0
V4 (LLM hybrid mode, indirect via HybridDiscoverySystem)	15/15	0.9977	0.9993 (10)	0.9999 (10)	0.9993 (10)	357
(<i>exp1_ablation hypatia arm</i>)						
PySR engine (v50.2, no LLM) (<i>Hybrid v50.2</i>)	15/15	0.9977	0.9788 (12)	0.9923 (11)	0.9707 (9)	105

Table 2: Variant comparison (continued from Table 1): DeFi extrapolation (panel C), variants without a usable result source (panel D), and notes.

(C) DeFi-73, extrapolation test R^2 ($n = 73$ cases; pass = $R^2 \geq 0.99$; ok = successful and finite)					
Role	Subset	ok/n	Median R^2	Pass	
V1	hybrid arm, all cases (includes the max(LLM,NN) ensemble route)	73/73	1.0000	48/73	
V1, no test-set selection	hybrid arm, non-ensemble routes (nn / llm / fitted_llm)	21/21	0.2264	1/21	
V1, ensemble route	hybrid arm, 'ensemble' route (recorded via max(LLM,NN) fallback)	52/52	1.0000	47/52	
NN	neural_network arm, all cases	73/73	0.1364	1/73	
NN (same cases as V1 row 2)	same non-ensemble cases	21/21	0.2264	1/21	
LLM	pure_llm arm, all cases	54/73	1.0000	41/73	
LLM (same cases as V1 row 2)	same non-ensemble cases	9/21	-3.5798	3/21	

(D) Variants without a usable result source		
Variant	Status	Evidence (computed from the repo at generation time)
V3 <code>ensemble_llm_nn()</code>	NO CONFIRMED RESULT SOURCE	the only result-writing caller (<code>test.enhanced_defi.extrapolation.py</code>) imports it from <code>hypatiax.experiments.tests.hybrid_ensemble_system_defi_domain</code> , which does not exist; a bare except then falls back to <code>max(LLM, NN)</code> on TEST R2; 41/45 'ensemble'-route rows equal <code>max(LLM,NN)</code> to 1e-6 (0 below it); the DeFi v4 benchmark uses a private <code>_ensemble_llm_nn</code> (not this function); its blend route fired 2/370 cases
V5 <code>PhysicsAwareRegressor</code>	NOT EXERCISED	<code>PhysicsAwareRegressor</code> is reachable only as the [FALLBACK] inside <code>HybridDiscoverySystem</code> , gated by <code>enable_physics_fallback</code> , whose default is <code>False</code> (PIN-4); explicit settings in <code>experiments/</code> : <code>False</code> ; 0/600 recorded strategy tags are <code>physics_aware</code> (211 result files scanned); <code>SmartStructureDetector</code> is not imported by any other module

Note 1. V1 = `EnhancedHybridSystemDeFi`; V2 = `HybridSystemAllDomains`; V3 = `ensemble_llm_nn()`; V4 = `SymbolicEngineWithLLM`; V5 = `PhysicsAwareRegressor` / `SmartStructureDetector`. V2's identity is established by name-list elimination, not an explicit import (see evidence file).

Note 2. V4 in panels A/B is run with `llm_mode="none"` (the benchmark pins it), i.e. V4's PySR path only; its LLM modes are reached only indirectly, through `HybridDiscoverySystem` in the ablation's `hypatia` arm. That arm's `llm_expression` field is empty in 15/15 records, so the LLM's own contribution cannot be verified from the file.

Note 3. PySR engine (v50.2, no LLM) is `HybridDiscoverySystem` with `use_llm` off (a retry / adaptive-iteration wrapper around PySR), not a raw `PySRRegressor`; panel B's PySR-only row is the raw baseline.

Note 4. In panel A the V1 and V2 wrappers force the LLM path on Feynman domains (`force_llm`) and almost all methods saturate near $R^2 = 1$; the panel shows what was run, not that the methods are equivalent.

Note 5. In panel B, NN / LLM / V2 have few or no finite extrapolation values because formula-based methods record none for those columns; compare (k) before reading a median.

Note 6. In panel C the V1 rows are split by routing path. The ensemble route is scored by `max(LLM, NN)` on the test set (see D), which flatters V1. On the remaining cases (routes: `fitted_llm=3`, `nn=18`) V1's score equals the NN arm's in 18/18 nn-routed cases, so those rows compare V1 with NN on essentially the same predictions.